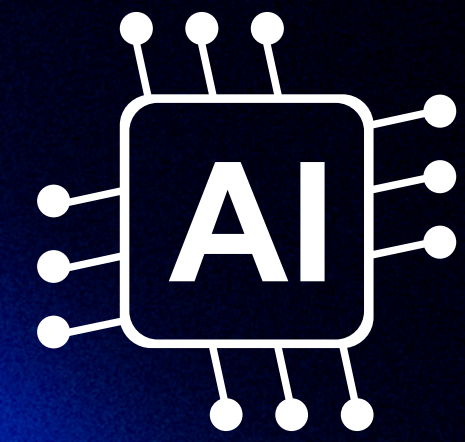




# AI-POWERED ORAL EXAMINATION PLATFORM

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Capstone Project  
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# BACKGROUND & MOTIVATION

- Rapid growth of generative AI (e.g., ChatGPT) challenges traditional written assessments
- Increased risk of academic dishonesty in take-home exams
- Oral exams are more reliable but:
  - Time-consuming
  - Not scalable
- Need for a scalable, automated, and fair assessment solution

**Goal:** Reimagine oral exams using AI

# PROJECT OBJECTIVES



- Define system scope, requirements, and architecture
- Develop a working prototype (MVP)
- Pilot and validate system performance
- Deliver deployment-ready platform and documentation

# WHAT WAS DONE

Designed for scalability +  
academic integrity

-  Built full-stack AI-powered oral exam system
-  Implemented voice interaction (STT + TTS)
-  Designed LLM-based grading system
-  Developed automated feedback pipeline

# SYSTEM IMPLEMENTATION

## Frontend (React)

- Exam interface
- Audio recording

## Backend (Python)

- API handling
- Data processing

## Database (SQLite)

- User & exam records

## AI Layer

- LLM-based scoring (ChatGPT)

Automated oral question delivery

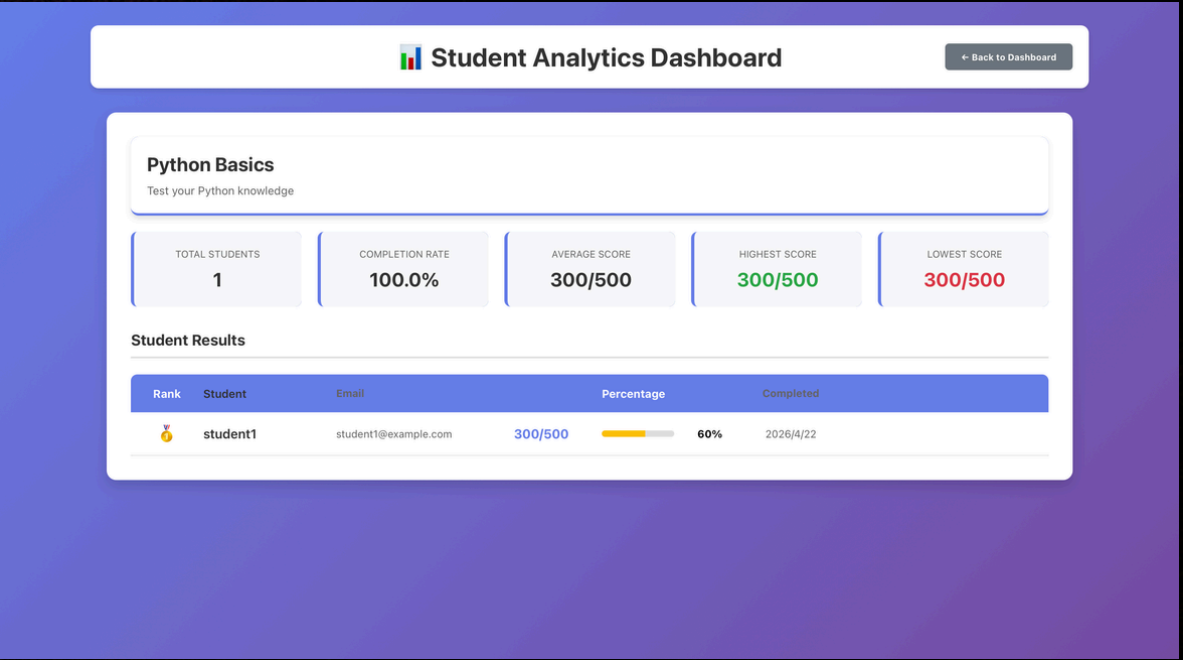
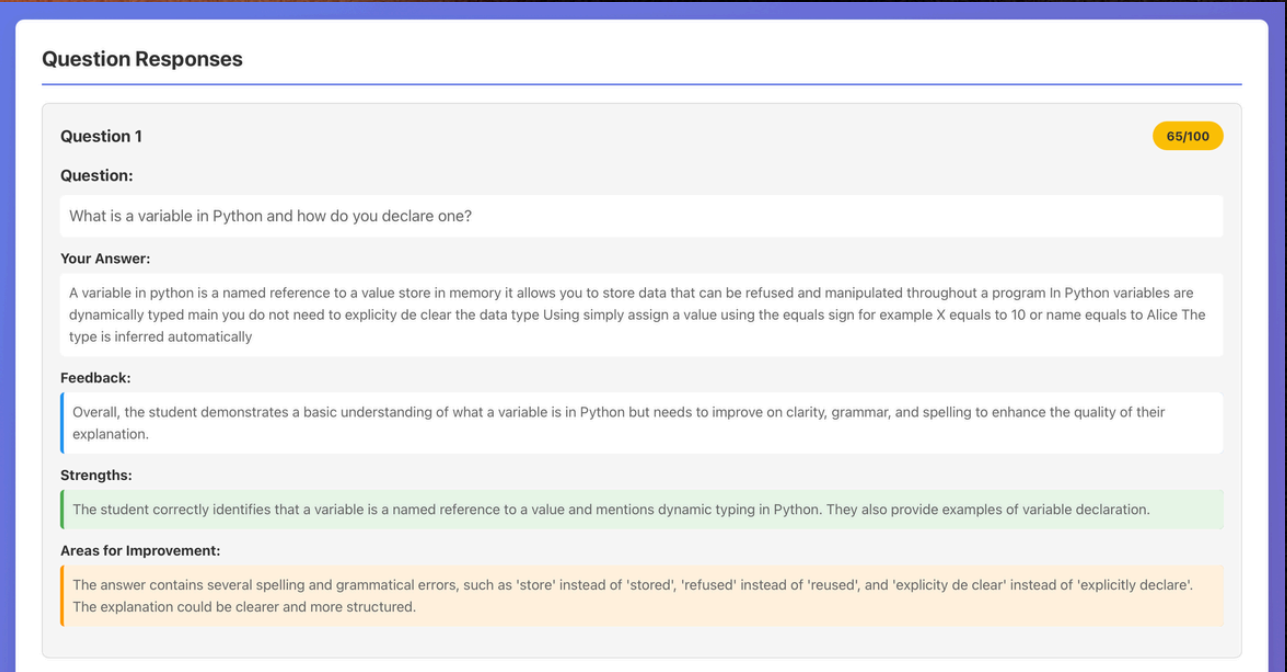
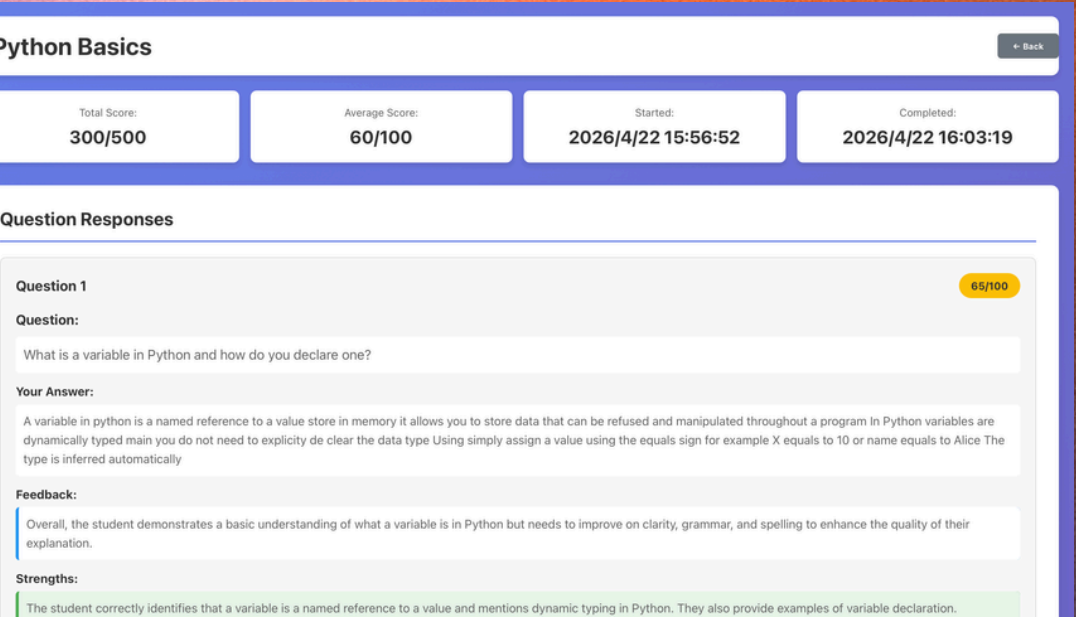
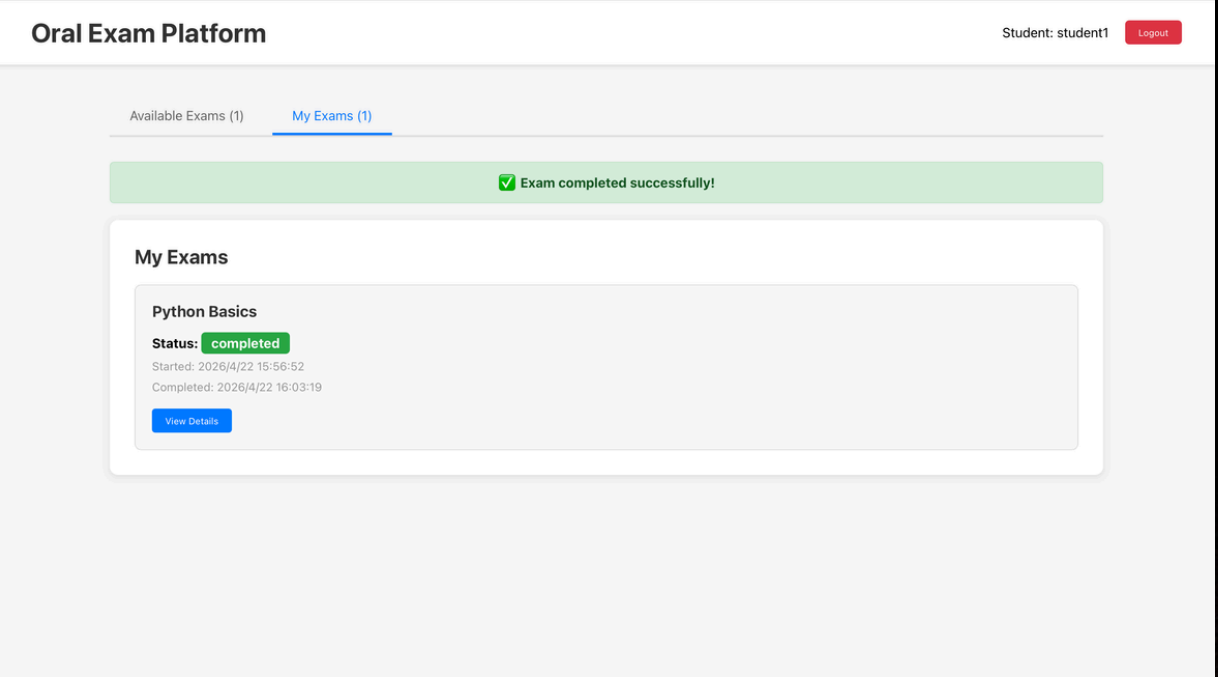
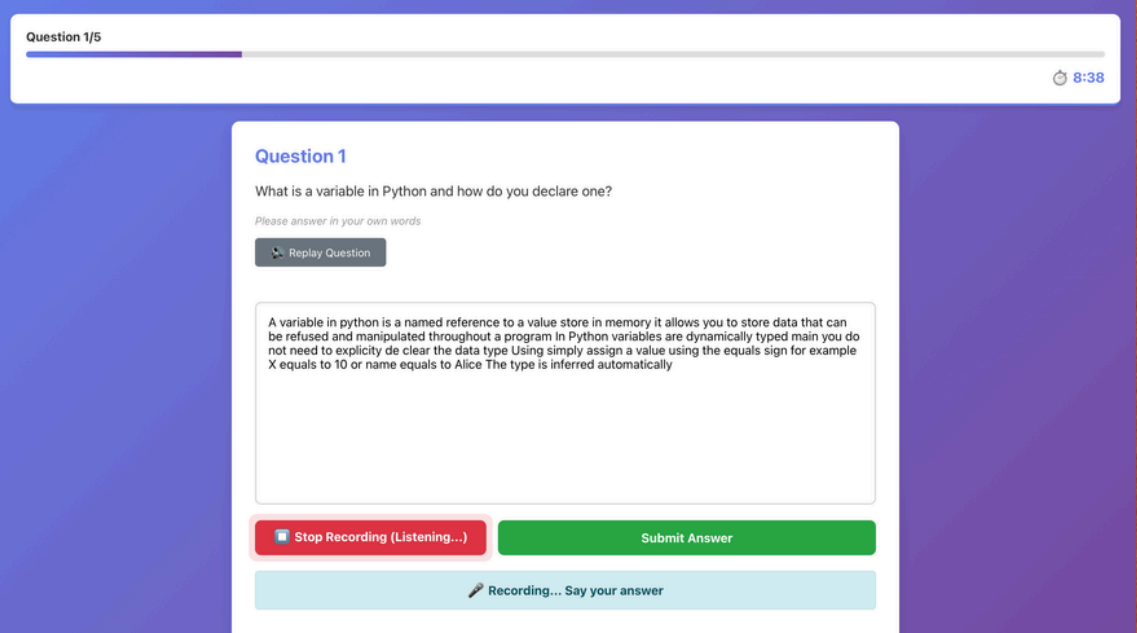
Real-time voice interaction

# SYSTEM CAPABILITIES

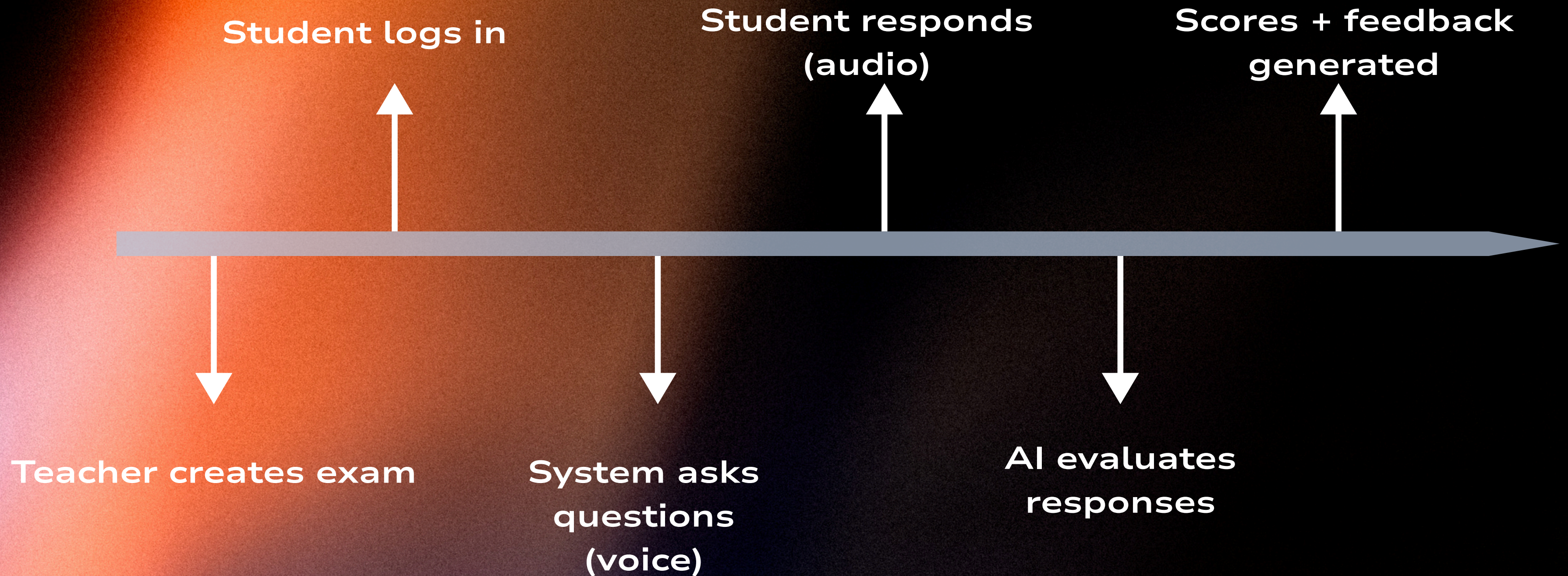
Multi-agent evaluation (LLM ensemble)

Personalized feedback

Teacher analytics dashboard



# PROCESS WORKFLOW



End-to-end automated oral exam

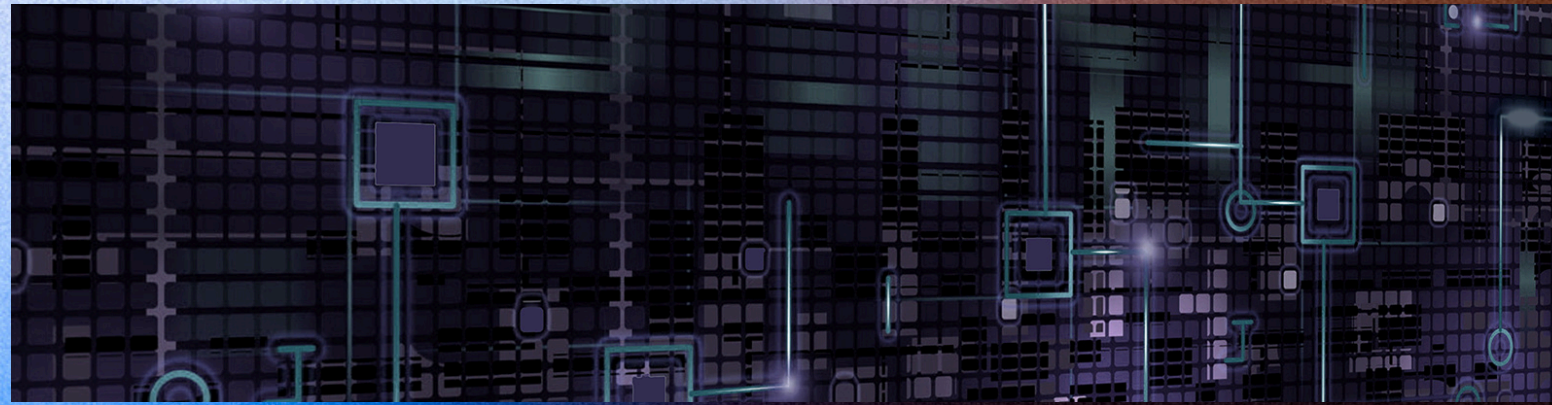
# APPROACH & METHODOLOGY

- Built full-stack prototype
- Designed simulated dataset:
  - 5 students × 5 questions
- Defined answer quality levels:
  - High / Medium / Low
- Conducted multiple evaluation runs  
Tested consistency and scoring reliability

|   | A                           | B          | C          | D          | E          | F          | G         | H          |
|---|-----------------------------|------------|------------|------------|------------|------------|-----------|------------|
| 1 | Student                     | Question 1 | Question 2 | Question 3 | Question 4 | Question 5 | Avg Score | Std Dev    |
| 2 | Student 1 (High)            | 95         | 90         | 90         | 90         | 90         | 91        | 2.23606798 |
| 3 | Student 2 (Medium)          | 90         | 85         | 80         | 80         | 80         | 83        | 4.47213595 |
| 4 | Student 3 (Low)             | 60         | 60         | 60         | 60         | 60         | 60        | 0          |
| 5 | Student 4 (Mixed: 2H/2M/1L) | 95         | 90         | 80         | 80         | 60         | 81        | 13.4164079 |
| 6 | Student 5 (Mixed: 1H/2M/2L) | 90         | 85         | 80         | 60         | 60         | 75        | 14.1421356 |

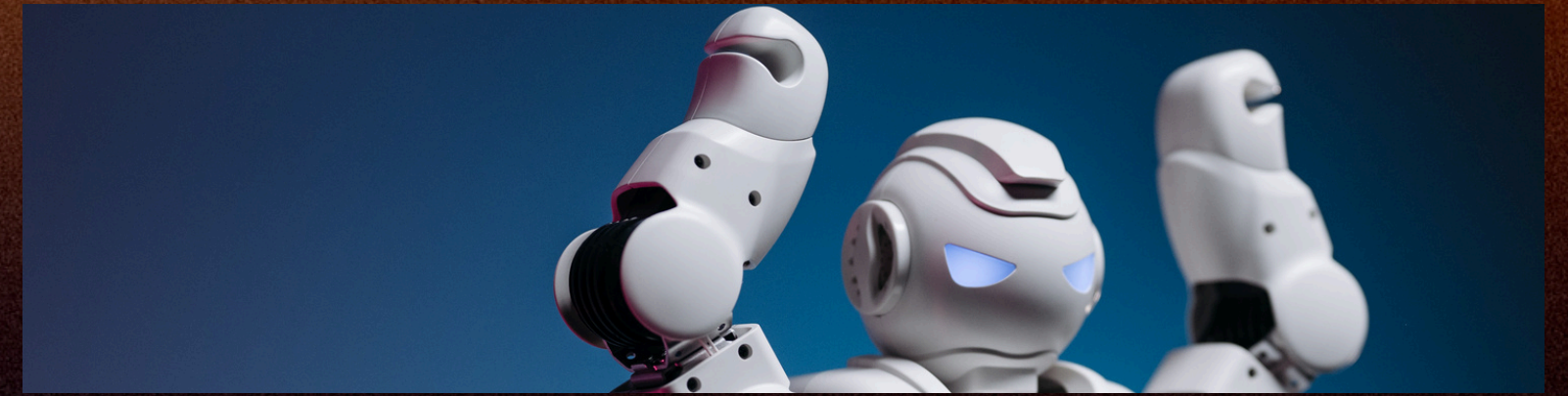
|   | A             | B        | C         | D         | E         |
|---|---------------|----------|-----------|-----------|-----------|
| 1 | Response Type | Question | Attempt 1 | Attempt 2 | Attempt 3 |
| 2 | High          | Q1       | 95        | 95        | 90        |
| 3 | Medium        | Q3       | 80        | 80        | 80        |
| 4 | Low           | Q5       | 60        | 60        | 60        |

# EVALUATION METRICS & RESULTS



## ➤ Evaluation Metrics

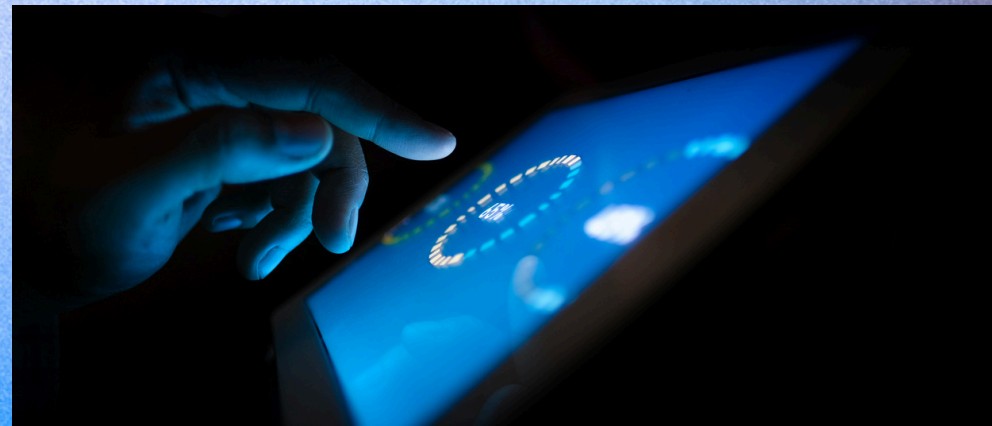
- **Scoring Consistency**  
→ Stability of AI evaluation across repeated runs
- **Evaluation Accuracy**  
→ Ability to distinguish High / Medium / Low quality responses
- **Variability (Std. Dev.)**  
→ Measure of scoring dispersion
- **System Usability**  
→ Ease of use for both students and instructors



## ➤ Results

- AI successfully differentiated High, Medium, and Low responses
- Average simulated scores improved (~60 → ~80)
- Repeated evaluations showed relatively stable scoring patterns
- Low standard deviation indicated acceptable consistency
- End-to-end workflow validated system feasibility

# PROJECT CHALLENGES & LIMITATIONS



## Challenges

- Variability in LLM-generated evaluations across runs
- Ensuring fairness and consistency in scoring
- Handling audio recording and processing reliability
- Integration complexity across frontend, backend, and AI models



## Lessons Learned

- Prompt design significantly impacts evaluation quality
- Multi-model (LLM ensemble) improves robustness
- Voice interaction enhances authenticity of assessment
- Modular system design supports scalability and flexibility



## Limitations

- Limited testing with simulated data rather than real students
- Dependence on LLM performance and potential bias
- Not fully optimized for production-level deployment
- Evaluation consistency still requires further validation

# CONCLUSION & RECOMMENDATIONS

## ➤ Conclusion

- Developed a scalable AI-powered oral examination platform
- Demonstrated feasibility of automated voice-based assessment
- Addressed academic integrity challenges in the generative AI era
- Showed potential for improving efficiency, scalability, and fairness in evaluation

## ➤ Future Work

- Conduct testing with real student data at larger scale
- Improve evaluation consistency and bias mitigation
- Integrate with Learning Management Systems (e.g., Brightspace)
- Enhance speech recognition and response analysis accuracy
- Deploy as a production-ready, cloud-based platform

# THANK YOU

**GitHub Repository:** <https://github.com/danxia2002/oral-exam-platform.git>

**Includes:**

- Source code
- Final report
- Testing data
- Presentation slides

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